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AXEL: High-pressure xenon gas TPC for neutrinoless double beta decay search

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ABSTRACT

A Xenon ElectroLuminescence (AXEL) is a high-pressure xenon gas time projection chamber with a unique ionization-electron counter called Electroluminescence Light Collection Cell (ELCC), which has been developed for neutrinoless double beta decay ($0\nu\beta\beta$) search. We have developed a dedicated Cockcroft–Walton high-voltage generator installed inside the chamber of a 180L-size prototype and applied up to −34.3 kV in xenon gas at 6.8 bar. A full-width at half-maximum energy resolution of (0.68 ± 0.01) % was obtained for 2615-keV electrons. Three-dimensional tracks of electrons were reconstructed, and the ability of the background rejection in $0\nu\beta\beta$ search is being evaluated. Toward the first $0\nu\beta\beta$ search with the AXEL detector, a 1000L-size detector is being constructed, and R&D of the components is in progress such as ELCC with large-area silicon photomultipliers, readout electronics, Cockcroft–Walton generator and scintillation detector.

1. Introduction

Observation of the neutrinoless double beta decay ($0\nu\beta\beta$) is crucial to reveal the Majorana nature of neutrinos. Since $0\nu\beta\beta$ would be extremely rare, e.g., the lower limit of the half-life time for ^{136}Xe is $T^{0\nu\beta\beta}_{1/2} > 3.8\times10^{26}$ years (90% C.L.) [1], a large number of double-beta-decay nuclei and background suppression are essential requirements for the search. For the large mass, scalability of the detector is important. To minimize background contamination, it is imperative to distinguish backgrounds by energy measurement with high resolution and also by other methods.

To meet these requirements, a high-pressure xenon gas time projection chamber (TPC) has been developed, initially proposed and demonstrated by the NEXT collaboration [2–4]. For the noble gas TPC, the electroluminescence (EL) process can be employed for ionization signal amplification. The EL process gives small amplification fluctuation because it is a linear process, thereby enabling high energy resolution. Furthermore, the track patterns reconstructed in three-dimensional space can be utilized to distinguish the $0\nu\beta\beta$ signal from backgrounds.

2. AXEL detector

A Xenon ElectroLuminescence (AXEL) detector is a high-pressure pure xenon gas TPC for $0\nu\beta\beta$ search shown in Fig. 1. Charged particles passing through the gas volume of the detector excite and ionize xenon atoms along their trajectories. Excited atoms subsequently emit primary scintillation photons, which are detected by scintillation detectors for the t_0 determination. Ionization electrons are drifted to an anode plane by an electric field of about 100 V/cm/bar and detected by our novel ionization-electron counter called Electroluminescence Light Collection Cell (ELCC).

The ELCC consists of 4 layers; an anode electrode, a PTFE plate with holes, a ground (GND) mesh electrode and silicon photomultiplier

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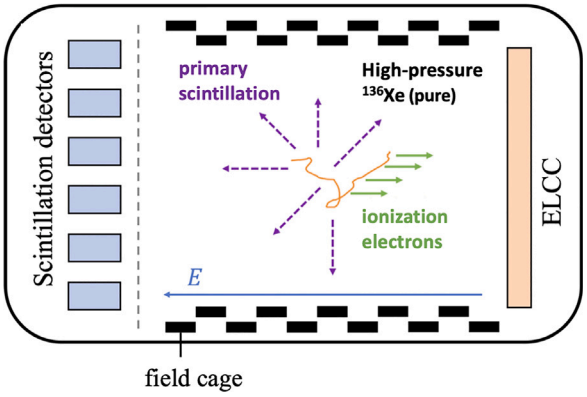


Fig. 1. Schematic drawing of the AXEL detector.

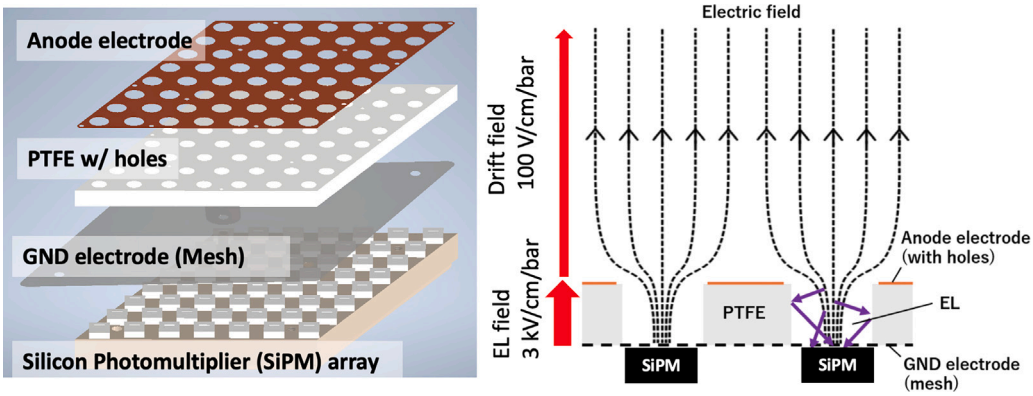


Fig. 2. Schematic drawing of ELCC structure (left) and principle of ELCC (right).

(SiPM) array, as illustrated in Fig. 2. Ionization electrons drifting toward the ELCC are collected into each hole by a stronger electric field (~3 kV/cm/bar) within the holes, causing the EL process. The emitted photons are detected by the SiPM located behind the ground mesh. The ELCC is composed of multiple units with identical structures, allowing for scalability. Furthermore, the pixelated structure of the ELCC enables the detection of arrival positions of ionization electrons, which are used to reconstruct three-dimensional tracks with timing information and to measure the energy deposited by the charged particle with excellent uniformity over the whole anode plane.

3. Updated performance demonstration of 180L-size prototype

To demonstrate great ability of the AXEL detector for $0\nu\beta\beta$ search, performance of the detector has been evaluated using a 180L-size prototype [5]. Components of the 180L-size prototype are shown in Fig. 3. ELCC plane has 672 channels and consists of 12 ELCC units. Each unit has 56 silicon photomultipliers, Hamamatsu multi-pixel photon counters (MPPCs) S13370-3050CN which is sensitive to xenon VUV EL photons. On the cathode side, 7 VUV-sensitive photomultiplier tubes (PMTs) are installed for scintillation detection. One of the new features is a high voltage generation by a Cockcroft–Walton generator installed inside the chamber. This eliminate the need for a high-pressure tolerable feedthrough for high voltage. We have achieved to apply −34.3 kV to cathode in xenon gas at 6.8 bar with 1.2 kV peak-to-peak AC input.

Measurements using a γ-ray source from ^{228}Th series with this setup was conducted over a period of approximately 2 months. During the measurement, the xenon gas pressure was maintained around 6.8 bar, and the electric field applied was 2.7 kV/cm/bar for the EL region and 90 V/cm/bar for the drift region.

The obtained EL photon count spectrum is shown in Fig. 4. The obtained full-width at half-maximum (FWHM) energy resolution at 2615 keV, which is close to the $0\nu\beta\beta$ Q-value of ^{136}Xe (2458 keV), is (0.67 ± 0.08) %. Using energy resolutions at various peaks in Fig. 4, FWHM energy resolution at the Q-value is estimated to be (0.68 ± 0.01) % to (0.79 ± 0.08) % as shown in Fig. 5. To further improve the energy resolution, contributions from various sources were estimated, revealing several non-negligible effects, such as mis-reconstruction of the position along the drift direction, statistical fluctuation of EL photon detection, and others. We are planning to address these effect through future detector upgrades.

Three-dimensional tracks of electrons were successfully reconstructed and are being used to evaluate a machine-learning based background rejection performance. Fig. 6 shows an example track deconvoluted with the Richardson–Lucy deconvolution method [6]. Clear tracks are obtained thanks to the deconvolution.

4. R&D for 1000L-size detector construction

To conduct the first $0\nu\beta\beta$ search using the AXEL detector and to demonstrate its excellent background rejection ability, a 1000L-size detector is being constructed. The detector is in approximately 1,000 m underground at the Kamioka Observatory to reduce cosmic-ray track accumulation. The chamber was installed in 2023, as shown in Fig. 7. R&D of the detector components is in progress, and its current status is described below.

ELCC unit

A new ELCC unit with 64 channels as seen in Fig. 8 is planned for use. This unit utilizes a round-shape MPPC (ϕ4.7 mm), which has approximately twice the detection area of the current one (3 mm square). The new MPPC covers the entire region of the ELCC holes

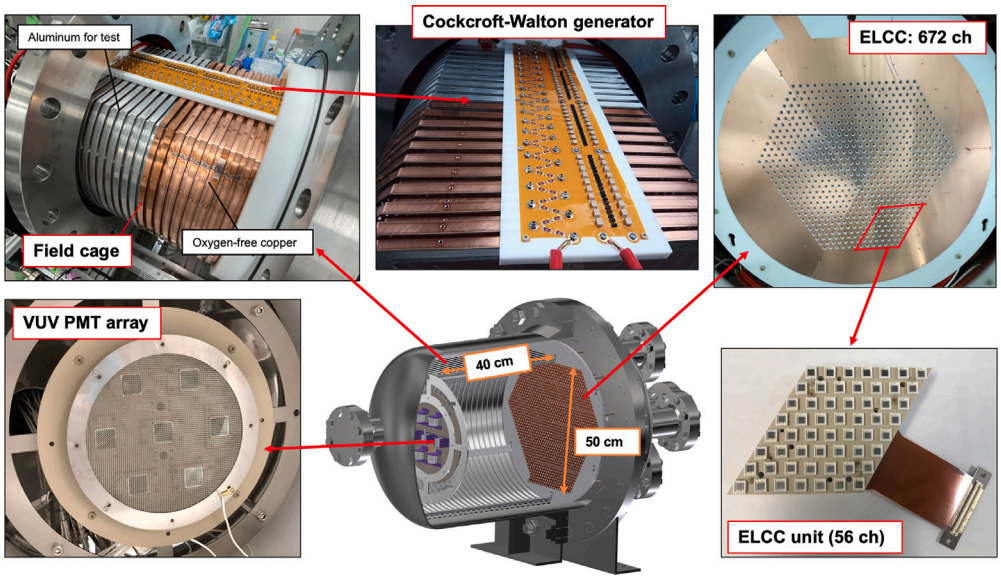


Fig. 3. 180L-size prototype components.

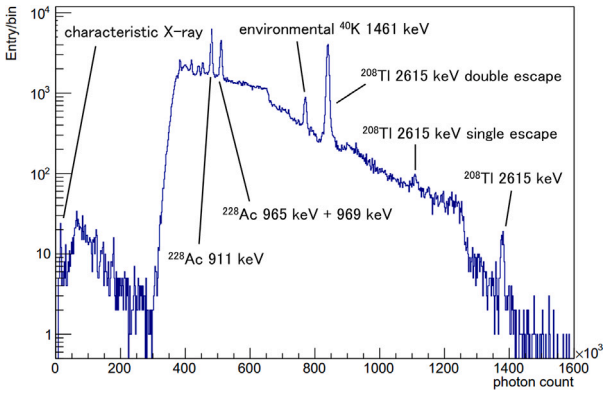


Fig. 4. Photon count spectrum. The low entries below 3×10^5 photon count correspond to the down sampling for low energy event.

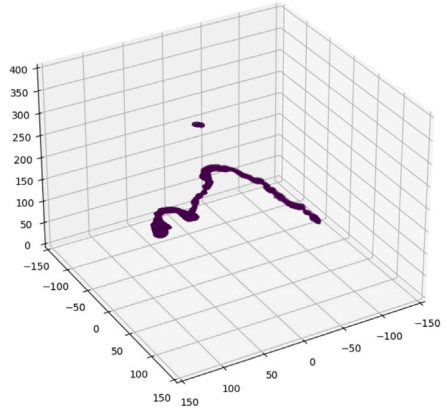


Fig. 6. Deconvoluted track of electrons of about 2615 keV. Short track on upper side is a photoelectron from characteristic X-ray. Long track at the center is a photoelectron from γ-ray of ^{208}Tl.

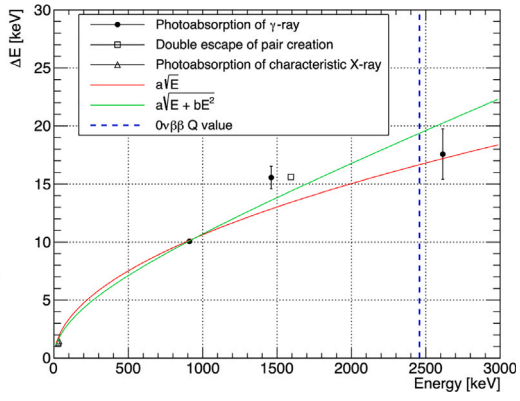


Fig. 5. Extrapolation of energy resolution to $0\nu\beta\beta$ Q-value. Photoabsorption of γ-ray peaks is used for the extrapolation. Red and green line represent the extrapolation results with the functions $a\sqrt{E}$ and $a\sqrt{E+bE^2}$, respectively.



Fig. 7. Chamber of the 1000L-size detector. Size of the chamber is 1.0 m diameter and 1.5 m length.

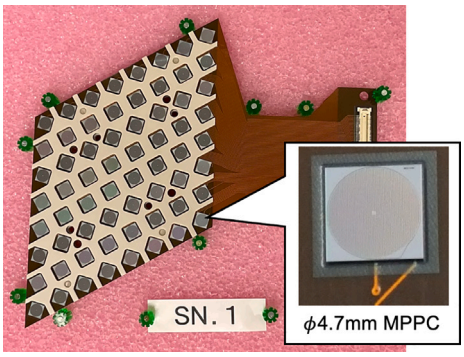


Fig. 8. ELCC unit with 64 channels. Round-shape (ϕ4.7 mm) MPPCs are mounted directory on flexible printed circuit.

(ϕ4.5 mm), so this modification is expected to increase the number of detected EL photons. The new MPPC is mounted directly on a flexible printed circuit to eliminate the ceramic package of the current one, which contains significant background sources.

Readout electronics

We have developed a dedicated readout electronics board for the EL signal [7]. The number of readout channels of the board was increased from 56 channels to 64 channels. The waveform is recorded at 5 MS/s. Different bias voltages can be applied to MPPCs. DC coupling is realized to avoid waveform distortion. The input impedance is assigned to be 10 Ω to suppress the MPPC-nonlinearity effect.

Cockcroft–walton generation

For the operation of the 1000L-size detector, it is necessary to generate −76.4 kV high voltage. To generate such high voltage with a Cockcroft–Walton generator, a high-power AC power supply was constructed using a commercial audio amplifier and a step-up transformer. With these components, −76 kV has been successfully generated in atmosphere at a peak-to-peak AC input of 1.8 kV. To achieve even higher voltage generation, a symmetrical Cockcroft–Walton generator [8] is under development to mitigate voltage drop.

Scintillation detection

The current scintillation detection efficiency of the PMTs installed in the 180L-size prototype is low due to the small detection area, resulting in the detection of only 0.7 photons per PMT at the Q-value. This limitation leads to mis-reconstruction of the position along the drift direction. To improve the efficiency, a scintillation detection plate has been developed, consisting of an acrylic plate with tetra-phenyl-butadiene (TPB) as a wavelength shifter coated on the surface, along with reflectors and MPPC (S13360-6075PE), as illustrated in Fig. 9. In this plate, scintillation photons are converted to visible one in the TPB layer and part of them are trapped within the acrylic layer through total reflection. The photons are then detected by the MPPC. This detector features much lower cost, about 4 times larger detection area and higher detection efficiency, thus it is expected to detect 11 photons per plate at the Q-value.

5. Summary

The AXEL project aims to search for $0\nu\beta\beta$ using a high-pressure xenon gas TPC equipped with a unique ionization-electron counter

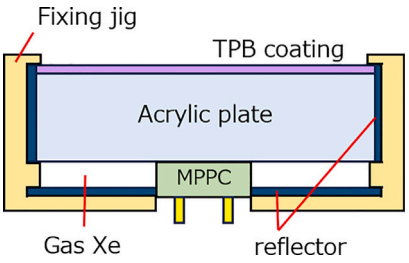


Fig. 9. Schematic drawing of scintillation detection plate. The plate consists of an acrylic plate coated with TPB on its surface, reflectors and MPPC.

called ELCC. The 180L-size prototype has demonstrated the excellent potential of the AXEL detector, showcasing its scalable structure, the ability to generate high voltage without a big feedthrough, excellent energy resolution, and the capability to reconstruct three-dimensional electron tracks. As a next step, various R&D efforts are underway for the 1000L-size detector, including an ELCC unit with 64 round-shaped MPPCs, readout electronics, the Cockcroft–Walton generator, and scintillation detectors, all aimed at the first $0\nu\beta\beta$ search with AXEL detector.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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